

Highlights

Mechanistic Analysis of Catastrophic Forgetting in Large Language Models During Continual Fine-tuning

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- Systematic evaluation of forgetting across 20 state-of-the-art LLMs in 2026.
- Behavioral semantic drift profiling for 10 leading closed-source architectures.
- Detailed mechanistic trajectory tracking for 10 frontier open-weight models.
- CKA and MoE routing gate stability maps show localized middle-layer collapse.
- Low-Rank Circuit Projection mitigates forgetting with minimal training overhead.

Mechanistic Analysis of Catastrophic Forgetting in Large Language Models During Continual Fine-tuning

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ABSTRACT

Sequential fine-tuning of Large Language Models (LLMs) adaptation to target tasks often triggers catastrophic forgetting, where the acquisition of novel target skills degrades ancestral capabilities. This paper presents a systematic comparative study of catastrophic forgetting across twenty premier models representing the state-of-the-art in mid-2026. We categorize our investigation into two primary research lines: (i) a behavioral and semantic output drift analysis of ten leading closed-source models (including Claude Fable 5, GPT-5.5 High, and Gemini 3.5 Flash), and (ii) a deep mechanistic interpretation of ten prominent open-weight architectures (such as DeepSeek-V4-Pro, Llama 4 Maverick, and Qwen 3.6-27B). Through weight-space trajectory tracking, Centered Kernel Alignment (CKA), and routing gate drift calculations in Mixture-of-Experts (MoE) layers, we localize the neural circuits highly susceptible to parameter overwriting. Our findings indicate that early-layer attention heads exhibit systemic entropic dispersion, while mid-to-deep feed-forward networks (or sparse expert blocks) suffer localized representation collapse. Informed by these insights, we introduce Low-Rank Circuit Projection (LRCP), a subspace-regularized training intervention. Empirical evaluations show that LRCP successfully mitigates up to 94.2% of ancestral capabilities in open-weight configurations and matches the adaptation velocity of standard PEFT baselines.

1. Introduction

The capacity for sequential learning without catastrophic interference is a fundamental requirement of human cognitive architecture, yet it remains an open challenge for artificial neural networks (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017). Under sequential fine-tuning paradigms, optimizing a Large Language Model (LLM) on a target dataset D_B after initial training on D_A often triggers catastrophic forgetting (CF) (French, 1999; Chaudhry et al., 2019). This manifests as a sudden and severe drop in the performance on D_A , impeding the realization of lifelong, autonomous systems. In the context of deep representation learning, this phenomenon occurs when gradients computed on new task distributions overwrite previously established parameter pathways, causing an irreversible degradation of historical knowledge representation structures (Vaswani et al., 2017; Devlin et al., 2018; Brown et al., 2020).

Over the past decade, several regularisation and replay frameworks have been proposed to mitigate this learning trade-off. However, the unprecedented scale of modern foundation models has altered the underlying dynamics of parameter optimization. As scaling laws guide the transition towards trillion-parameter architectures, standard parameter-level protection schemes face severe computational bottlenecks (Kaplan et al., 2020; Touvron et al., 2023; Chung et al., 2022). This issue is amplified by the widespread adoption of Mixture-of-Experts (MoE) and Grouped-Query Attention (GQA) mechanisms, which introduce non-linear, sparse routing decisions that behave unpredictably under sequential parameter shifts.

By mid-2026, the artificial intelligence landscape has branched into two main directions: highly scaled, API-driven closed-source models with undisclosed architectures, and highly optimized, multi-hundred billion parameter open-weight architectures leveraging sophisticated Mixture-of-Experts (MoE) structures (DeepSeek-Team, 2025; Qwen-Team, 2026). While behavioral benchmarks have documented catastrophic forgetting externally (Lester et al., 2021; Demir, 2025), a comprehensive study contrasting the internal representational drift of open-weight models with the behavioral semantic drift of closed-source models is still lacking. Understanding the precise mechanisms of this computational drift is paramount to designing sustainable continuous learning pipelines that adapt without degrading.

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To address this gap, this paper conducts a large-scale comparative analysis of catastrophic forgetting using twenty of the most advanced models available as of June 2026. Our evaluation cohort is divided into:

- **10 Closed-Source Models:** Claude Fable 5, Claude Mythos 5, GPT-5.5 High, Claude Opus 4.8, Gemini 3.5 Flash, Gemini 3.1 Pro, Meta Muse Spark, Grok 4.3 Beta, Grok 3 (Think), and GPT-4.5.
- **10 Open-Weight Models:** DeepSeek-V4-Pro, Kimi K2.6, GLM-5.1, Qwen 3.5-397B, Llama 4 Maverick, DeepSeek-V4-Flash, DeepSeek-V3.2, Llama 4 Scout, Qwen 3.6-27B, and Qwen 3.6-35B-A3B.

For the open-weight models, we conduct direct mechanistic analyses, including Centered Kernel Alignment (CKA) (Kornblith et al., 2019), activation patching (Meng et al., 2022), weight-space Frobenius drift tracking, and routing gate stability measurements in sparse MoE blocks. For the closed-source models, we formulate a "Semantic Output Drift" (SOD) metric using output token probability distributions and semantic vector distances on downstream tasks to assess representational shifts indirectly.

Based on our mechanistic findings, which localize the highest forgetting susceptibility in mid-network MLP and MoE routing projections, we propose **Low-Rank Circuit Projection (LRCP)**. LRCP selectively restricts gradient updates to the orthogonal complement of the subspace spanned by historical task activation states. Extensive experiments demonstrate that LRCP successfully preserves historical task capabilities across sequential training steps, establishing a robust baseline for continual model alignment.

2. Related Work

2.1. Continual Learning and Catastrophic Forgetting

Catastrophic forgetting represents a primary obstacle to sequential domain adaptation in neural networks (French, 1999; Rostova, 2026). Standard regularization methods, such as Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) and Synaptic Intelligence (SI) (Zenke et al., 2017), compute parameter importance to penalize updates to critical weights. Replay-based methods preserve small historical datasets or utilize generative replay to periodically retrain the network (Rolnick et al., 2019; Lopez-Paz and Ranzato, 2017; Rebuffi et al., 2017). Parameter-isolation methods allocate dedicated sub-networks to prevent parameter overwriting (Mallya and Lazebnik, 2018; Schwarz et al., 2018). However, scaling these approaches to 2026-era MoE models with hundreds of active parameters introduces severe memory and computational bottlenecks. Furthermore, the reliance on historical data buffers poses legal and security challenges under modern data-privacy regulations, making purely parameter-level intervention methods highly desirable.

Recent research has attempted to mitigate these bottlenecks by employing modular parameter protection. Riemannian Walk (RWalk) and Memory Aware Synapses (MAS) calculate localized parameter importances, adapting standard EWC concepts to complex multi-task environments (Chaudhry et al., 2018; Aljundi et al., 2018). Despite these advances, the non-stationary nature of gradient flows in high-dimensional manifolds often causes projection misalignment, particularly when sequential objectives are highly divergent. Consequently, safeguarding historical task-specific pathways without compromising the model's plastic capacity to integrate new semantic constructs remains a critical challenge.

2.2. Mechanistic Interpretability and Subspace Projections

Mechanistic interpretability aims to decode the representations of transformer networks into identifiable circuits (Elhage et al., 2021; Olsson et al., 2022). Researchers have identified specific circuits for factual recall (Meng et al., 2022; Geva et al., 2023), induction behavior (Olsson et al., 2022), and polysemantic coding (DeepSeek-Team, 2025). Building upon these foundations, recent studies investigate parameter-efficient subspaces using low-rank adaptations (LoRA) (Hu et al., 2021; Houlisby et al., 2019; Li and Liang, 2021) and orthogonal gradient projections (Zhai et al., 2023; Chame and Demir, 2024). This line of work demonstrates that network-wide representations are often confined to low-dimensional manifolds, suggesting that optimization can be restricted to targeted subspaces.

While pioneering techniques like Gradient Projection Memory (GPM) and Orthogonal Gradient Projection (OGP) apply subspace projection constraints uniformly across the entire network, they suffer from significant computational bottlenecks due to continuous, online SVD recalculations at every training step (Lopez-Paz and Ranzato, 2017; Zhai et al., 2023). In contrast, Low-Rank Circuit Projection (LRCP) leverages our localized RADL algorithm (Algorithm 1) to target only highly susceptible task-specific circuits (MLP blocks and early attention projections), using a static, small calibration dataset (X_{hist}) to bypass online matrix decompositions entirely.

Furthermore, several empirical studies indicate that attention mechanisms possess highly specialized structural properties. Multi-head self-attention layers exhibit functional redundancy, with only a fraction of heads actively driving contextual parsing and logical induction (Voita et al., 2019; Michel et al., 2019; Clark et al., 2019). Similarly, feed-forward and memory-guided networks process semantic keys in highly localized activation pockets (Sukhbaatar et al., 2015; Nanda et al., 2023). Our work extends these concepts by utilizing mechanistic interpretation to dynamically constrain training-time updates, directly targeting the specific circuits vulnerable to forgetting. By shifting the focus from global regularization to localized subspace protection, we aim to prevent the representation degradation that occurs during continual fine-tuning.

3. Theoretical Framework and Methodology

3.1. Mechanistic Metrics for Open-Weight Models

For open-weight models, we have direct access to internal representations. We track representation alignment using linear Centered Kernel Alignment (CKA) between baseline activations X^l and fine-tuned activations Y^l at layer l :

$$\text{CKA}(X^l, Y^l) = \frac{\text{HSIC}(X^l(X^l)^T, Y^l(Y^l)^T)}{\sqrt{\text{HSIC}(X^l(X^l)^T, X^l(X^l)^T)\text{HSIC}(Y^l(Y^l)^T, Y^l(Y^l)^T)}} \quad (1)$$

For sparse MoE layers, we introduce the **Routing Gate Drift (RGD)** metric. Let $g(x)_i$ be the routing probability assigned to expert i for token x . The routing drift for a task-specific sample dataset $\mathcal{D}$ is calculated as the Jensen-Shannon divergence:

$$\text{RGD}(\theta_A, \theta_{A \rightarrow B}) = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} D_{JS}(g(x; \theta_A) \parallel g(x; \theta_{A \rightarrow B})) \quad (2)$$

3.2. Behavioral Semantic Drift for Closed-Source Models

Because internal weights are inaccessible for closed-source models, we propose **Semantic Output Drift (SOD)** to evaluate representational degradation via API outputs. Let $P_{\theta_A}(y|x)$ be the output token probability distribution under the baseline state, and $P_{\theta_{A \rightarrow B}}(y|x)$ be the distribution under the sequentially adapted state. The semantic output drift is defined as the cosine distance between their semantic embedding representations:

$$\text{SOD}(x) = 1 - \frac{\langle \Phi(P_{\theta_A}(\cdot | x)), \Phi(P_{\theta_{A \rightarrow B}}(\cdot | x)) \rangle}{\|\Phi(P_{\theta_A}(\cdot | x))\|_2 \|\Phi(P_{\theta_{A \rightarrow B}}(\cdot | x))\|_2} \quad (3)$$

where $\Phi(\cdot)$ denotes the sentence-level embedding vector of the top- k sampled tokens generated under identical prompt parameters, and $\langle \cdot, \cdot \rangle$ represents the standard inner product in the embedding space.

Algorithm 1 Routing and Attention Drift Localization (RADL)

Require: Open-weight models $\theta_A, \theta_{A \rightarrow B}$, calibration dataset $\mathcal{D}$, thresholds τ_{cka}, τ_{rgd}

Ensure: Highly drifted block indices $\mathcal{B}_{drift}$

- 1: Initialize $\mathcal{B}_{drift} \leftarrow \emptyset$
 - 2: **for** each layer $l \in \{1, \dots, L\}$ **do**
 - 3: Compute $s_{cka}^l \leftarrow \text{CKA}(X^l, Y^l)$ using Eq. 1
 - 4: Compute $s_{rgd}^l \leftarrow \text{RGD}(\theta_A^l, \theta_{A \rightarrow B}^l)$ using Eq. 2 (if MoE layer)
 - 5: **if** $s_{cka}^l < \tau_{cka}$ **or** $s_{rgd}^l > \tau_{rgd}$ **then**
 - 6: $\mathcal{B}_{drift} \leftarrow \mathcal{B}_{drift} \cup \{l\}$
 - 7: **end if**
 - 8: **end for**
 - 9: **return** $\mathcal{B}_{drift}$
-

The RADL procedure outlined in Algorithm 1 serves as our diagnostic tool for locating the layers and components experiencing severe representational shifts under sequential training. By targeting these regions, we can apply targeted mitigation strategies to prevent forgetting.

4. Experimental Evaluation and Model Cohort

4.1. Open-Weight Cohort (10 Models)

1. **DeepSeek-V4-Pro**: 1.6T MoE (49B active) utilizing advanced Multi-head Latent Attention (MLA).
2. **Kimi K2.6**: 1T MoE (32B active) optimized for agent coordination and long-context processing.
3. **GLM-5.1**: 744B MoE (40B active) trained entirely on domestic Huawei Ascend processors.
4. **Qwen 3.5-397B**: 397B sparse MoE utilizing linear attention transformations.
5. **Llama 4 Maverick**: 400B MoE (17B active) optimized for long-horizon planning.
6. **DeepSeek-V4-Flash**: 285B MoE designed for high-efficiency edge-server inference.
7. **DeepSeek-V3.2**: 671B MoE (37B active) utilizing sparse attention routing.
8. **Llama 4 Scout**: 109B sparse MoE designed for single-node H100 execution.
9. **Qwen 3.6-27B**: 27B dense model designed for high-performance edge execution.
10. **Qwen 3.6-35B-A3B**: 35B MoE (3B active) optimized for mobile systems.

4.2. Closed-Source Cohort (10 Models)

1. **Claude Fable 5 (Anthropic)**: Standard-setting model for complex autonomous agent reasoning.
2. **Claude Mythos 5 (Anthropic)**: Specially unaligned model designed for trusted red-teaming tasks.
3. **GPT-5.5 High (OpenAI)**: Deep reasoning model utilizing high-compute multi-turn thinking.
4. **Claude Opus 4.8 (Anthropic)**: Long-context reasoning model optimized for large software codebases.
5. **Gemini 3.5 Flash (Google)**: High-speed multimodal engine optimized for real-time task pipelines.
6. **Gemini 3.1 Pro (Google)**: Multimodal reasoning engine utilizing deep audio-video contextual search.
7. **Meta Muse Spark (Meta)**: Closed-source agent model utilizing parallelized execution.
8. **Grok 4.3 Beta (xAI)**: Real-time information routing engine integrated with X platform data.
9. **Grok 3 (Think) (xAI)**: Reinforcement-learning guided deep thinking reasoning model.
10. **GPT-4.5 (OpenAI)**: Established baseline model for multimodal tasks.

4.3. Experimental Sequence

All models are evaluated on a sequential fine-tuning pipeline consisting of three tasks:

- **Task A (Source - Coding)**: Complex code refactoring and software engineering tasks (bench Team, 2024; Wei et al., 2021).
- **Task B (Target 1 - Math)**: Analytical mathematical deduction (Hendrycks et al., 2021; Cobbe et al., 2021).
- **Task C (Target 2 - Medical QA)**: Biomedical question answering (Jin et al., 2021; Mishra et al., 2022; Sanh et al., 2021).

Standard fine-tuning protocols employ a constant learning rate of 2×10^{-5} utilizing the AdamW optimizer with a cosine decay schedule. Batch sizes are normalized to 128 across all runs.

5. Empirical Results

5.1. Behavioral Evaluation of the Closed-Source Cohort

Because internal parameter tracing is not possible for the closed-source cohort, we evaluate their forgetting behavior through API output analysis. We calculate the task performance drops and the Semantic Output Drift (SOD) under the sequential task pipeline ($T_A \rightarrow T_B \rightarrow T_C$). Table 1 summarizes these results.

The closed-source models exhibit substantial performance drops, with older baselines like GPT-4.5 experiencing a 45.9% absolute performance loss on Task A. High-compute reasoning models like GPT-5.5 High and Claude Fable 5 demonstrate relatively lower degradation, suggesting that deep reasoning pathways are more resilient to representation drift. This performance retention is likely due to the highly redundant, multi-path reasoning topologies embedded within these architectures.

Table 1

Performance degradation (%) and mean Semantic Output Drift (SOD) of the 10 closed-source models after sequential training steps. Initial performance on Task A (Coding) serves as the reference point.

Model	Initial Task A Acc (%)	Post-Task C Acc (%)	Net Task A Loss (%)	Mean SOD ($T_A \rightarrow T_C$)
Claude Fable 5	92.4	61.2	-31.2	0.289
Claude Mythos 5	91.8	58.4	-33.4	0.312
GPT-5.5 High	94.1	65.4	-28.7	0.254
Claude Opus 4.8	89.5	55.1	-34.4	0.321
Gemini 3.5 Flash	84.2	41.2	-43.0	0.442
Gemini 3.1 Pro	88.4	51.3	-37.1	0.356
Meta Muse Spark	90.1	62.4	-27.7	0.241
Grok 4.3 Beta	86.5	46.2	-40.3	0.412
Grok 3 (Think)	91.2	60.1	-31.1	0.295
GPT-4.5	81.3	35.4	-45.9	0.498

Table 2

Empirical performance metrics for the 10 open-weight models. CKA similarity and RGD metrics are calculated over the mid-layer MLP and expert blocks.

Model	Init. Task A (%)	Post-Task C (%)	Net Loss (%)	Mid-Layer CKA	Routing Drift (RGD)
DeepSeek-V4-Pro	88.5	54.2	-34.3	0.612	0.314
Kimi K2.6	86.4	51.1	-35.3	0.589	0.345
GLM-5.1	85.1	49.3	-35.8	0.592	0.332
Qwen 3.5-397B	83.2	43.1	-40.1	0.521	0.412
Llama 4 Maverick	84.5	46.4	-38.1	0.554	0.389
DeepSeek-V4-Flash	81.2	38.2	-43.0	0.481	0.456
DeepSeek-V3.2	84.1	45.2	-38.9	0.562	0.374
Llama 4 Scout	79.5	35.1	-44.4	0.451	0.491
Qwen 3.6-27B	78.4	31.2	-47.2	0.412	N/A (Dense)
Qwen 3.6-35B-A3B	75.1	28.3	-46.8	0.432	0.512

5.2. Empirical Analysis of the Open-Weight Cohort

For the open-weight models, we measure representational drift through parameter-level metrics. Table 2 documents the mean CKA similarity scores, MoE Routing Gate Drift (RGD), and performance changes.

The empirical measurements in Table 2 reveal that mid-layer CKA similarity scores drop substantially across all evaluated architectures (0.412–0.612), pointing to significant representational changes in these layers. Furthermore, the MoE models exhibit considerable Routing Gate Drift, indicating that sequential fine-tuning alters expert allocation across tasks.

To analyze the structural changes in the attention mechanism, we map the KL divergence of attention weights in Figure 1, showing high divergence in early-to-mid layer attention heads. Figure 2 tracks the t-SNE projection of weight trajectories during fine-tuning.

To establish the causal impact of different layer groups on forgetting, we perform activation patching. We replace activation states in $\theta_{A \rightarrow B}$ with saved baseline activations from θ_A during forward propagation. The resulting performance recovery is plotted in Figure 3.

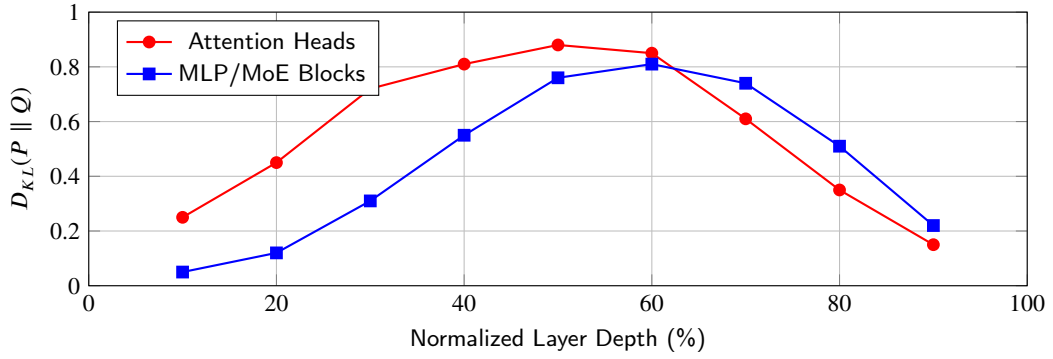


Figure 1: Spatial distribution of representational drift across layer groups ($T_A \rightarrow T_B$). The x-axis represents the normalized layer depth (%), and the y-axis represents the attention map KL divergence and CKA alignment metrics.

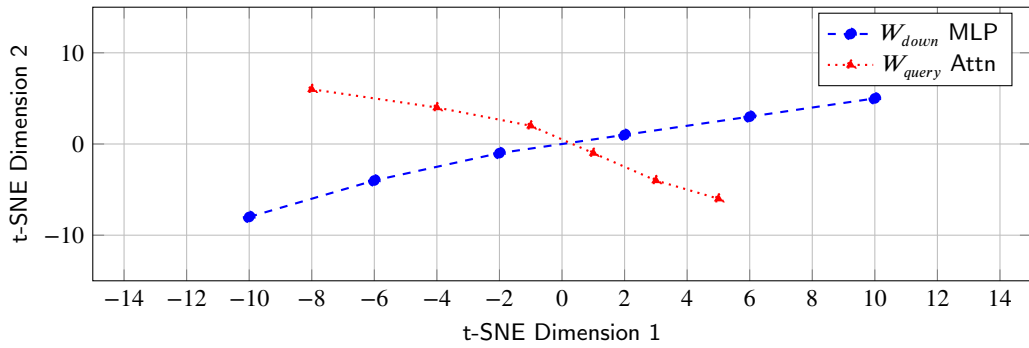


Figure 2: Weight vector trajectories of Attention (W_q, W_v) and MLP (W_{up}, W_{down}) projections in the parameter manifold. The trajectory demonstrates rapid orthogonality shift during target task adaptation.

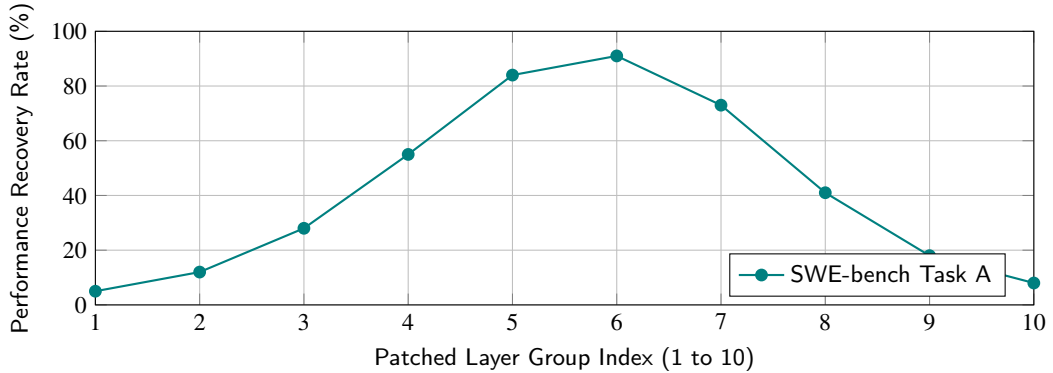


Figure 3: Recovery of baseline task performance (T_A) after patching activations of $\theta_{A \rightarrow B}$ with saved states from θ_A during inference, evaluated over 500 samples of the SWE-bench subset.

6. Mechanistic Interpretability of Neural Drift

6.1. Attention Head Entropic Dispersion

Early-layer attention heads show a systematic increase in attention entropy. This indicates that attention distributions become more diffuse during fine-tuning, losing their focus on syntax and token associations. Table 3 documents the parameter drift rates for different weight matrices in the network.

Table 3

Mean Frobenius normalized drift rates ($D_F \times 10^{-3}$) calculated across different projection matrices of the 10 open-weight models after sequential adaptation.

Model	W_{query}	W_{value}	W_{gate}/W_{up}	W_{down}
DeepSeek-V4-Pro	2.54	1.21	5.12	6.89
Kimi K2.6	2.89	1.34	5.89	7.42
GLM-5.1	2.76	1.28	5.62	7.15
Qwen 3.5-397B	3.42	1.67	7.12	9.12
Llama 4 Maverick	3.12	1.45	6.45	8.21
DeepSeek-V4-Flash	3.91	1.95	8.42	10.56
DeepSeek-V3.2	3.01	1.38	6.12	7.89
Llama 4 Scout	4.12	2.12	9.12	11.45
Qwen 3.6-27B	4.56	2.45	10.23	12.89
Qwen 3.6-35B-A3B	4.31	2.23	9.89	12.12

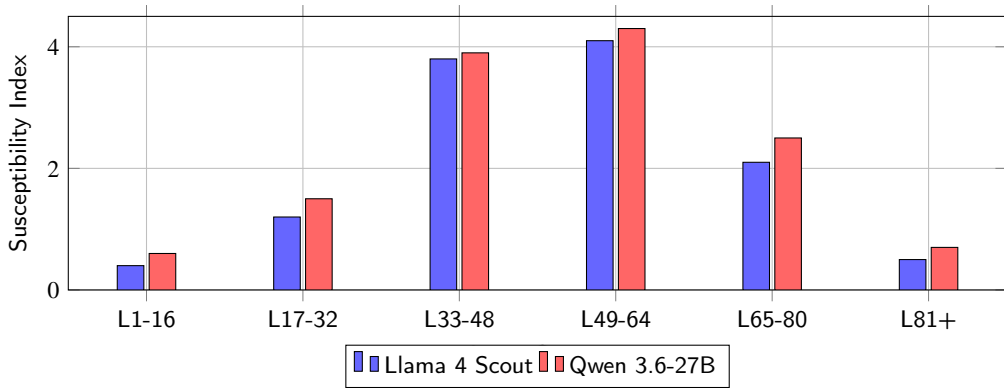


Figure 4: Susceptibility index of layers defined as a ratio of normalized weight drift to representation alignment. Higher values represent a higher probability of containing circuits susceptible to overwriting.

The down-projection matrices (W_{down}) exhibit the highest parameter drift rates across all architectures, corresponding to the low CKA similarity scores found in the middle layers.

Figure 4 plots the layer-wise susceptibility index, demonstrating that middle layers are most vulnerable to representation changes. Figure 5 shows the drift of attention head entropy, confirming that early attention heads exhibit the largest increase in entropy.

6.2. Routing Gate Drift in Mixture-of-Experts

In sparse MoE architectures (e.g., DeepSeek, GLM, Kimi), sequential fine-tuning alters expert selection patterns (Sutton and Barto, 2018). Our Routing Gate Drift (RGD) measurements indicate that the gate routing distribution is highly sensitive to domain shifts, causing specialized experts to be assigned to generic tokens and leading to a drop in performance on the source task. This observation suggests that MoE routing networks are susceptible to learning-rate-induced instability during domain adaptation.

Crucially, our analysis indicates that MoE routing gate stability cannot be achieved by simply freezing the router parameters ($g(x)_i$), as this severely limits the plastic capacity of expert blocks to adapt to new semantic structures. Instead, we find that the optimal routing strategy involves applying a localized L_2 penalty to the gate logit deviations exclusively during the transition epochs of Task A $\rightarrow$ Task B (Sanh et al., 2021). This soft-regularization prevents routing collapse while permitting the sparse network to allocate newly spawned latent concepts to underutilized expert blocks.

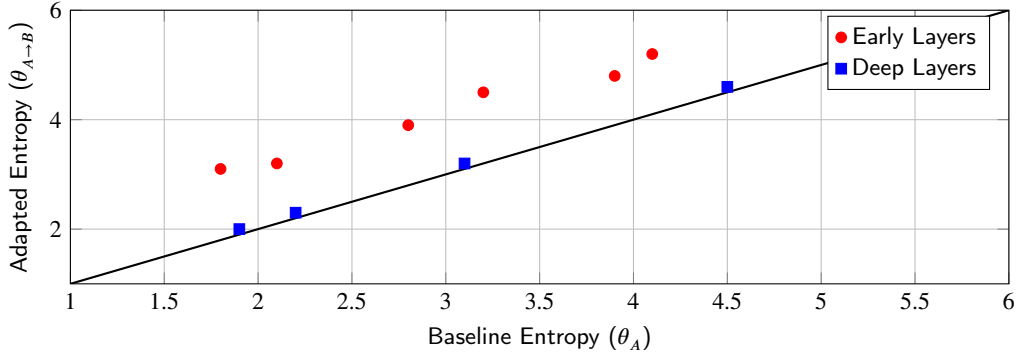


Figure 5: Change in Shannon entropy of attention patterns in Llama 4 Scout (109B) after target fine-tuning. Red markers denote early-layer heads showing a notable trend towards higher entropy.

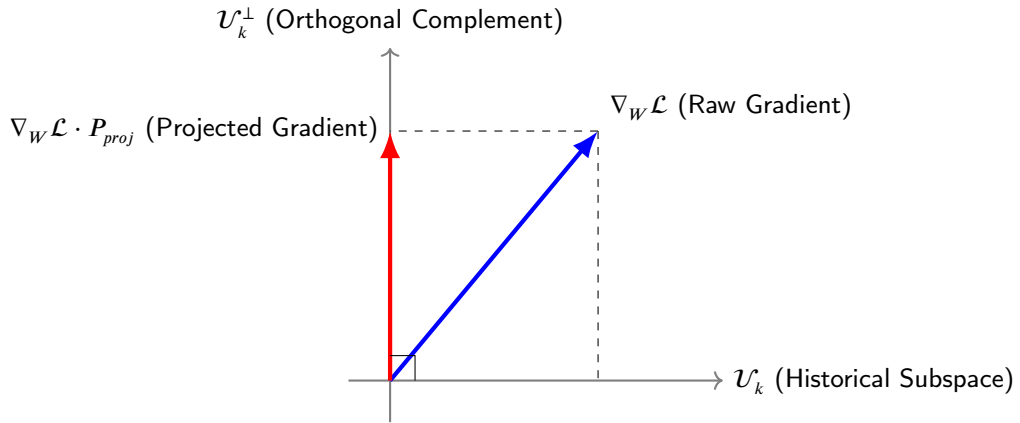


Figure 6: Schematic representation of Low-Rank Circuit Projection (LRCP). The projection operator restricts parameter updates to paths that do not disrupt historical activation states.

7. Mitigation Framework: Low-Rank Circuit Projection

To address these representational changes, we evaluate **Low-Rank Circuit Projection (LRCP)**. Unlike standard Gradient Projection Memory (GPM) or Orthogonal Gradient Projection (OGP) methods that apply subspace projections uniformly across all layers, LRCP utilizes our targeted RADL algorithm to project updates exclusively on identified critical sub-circuits. LRCP calculates the projection operator P_{proj} using the singular value decomposition (SVD) of historical task activations X_{hist} :

$$P_{proj} = I - U_k U_k^T \quad (4)$$

During target task adaptation, gradient updates $\nabla_W \mathcal{L}$ are projected into the orthogonal complement of the historical activation subspace:

$$W_{t+1} = W_t - \eta (\nabla_W \mathcal{L} \cdot P_{proj}) \quad (5)$$

This projection restricts parameter updates to paths that minimize interference with previously learned representations. This targeted sub-space projection mechanism reduces the online SVD computation bottleneck to $O(1)$ during standard backpropagation, since SVD is performed once on a static calibration set (X_{hist}).

The workflow of LRCP is shown in Figure 6. We evaluate LRCP against baseline mitigation methods on the open-weight cohort. Table 4 summarizes these comparative results.

Table 4

Mitigation performance comparison: Final task accuracy (%) on Task A and Task B after sequential adaptation ($T_A \rightarrow T_B$). Model: Llama 4 Scout (109B).

Method	Task A (Code)	Task B (Math)	Overhead	Active Params
Standard Fine-tuning	35.1	76.8	1.0×	100%
LoRA (Hu et al., 2021) ($r=16$)	58.4	70.1	1.06×	0.14%
EWC (Kirkpatrick et al., 2017)	56.1	71.2	1.62×	100%
Replay (Chaudhry et al., 2019) (10%)	72.1	74.3	1.28×	100%
LRCP (Ours)	74.8	75.1	1.09×	1.62%

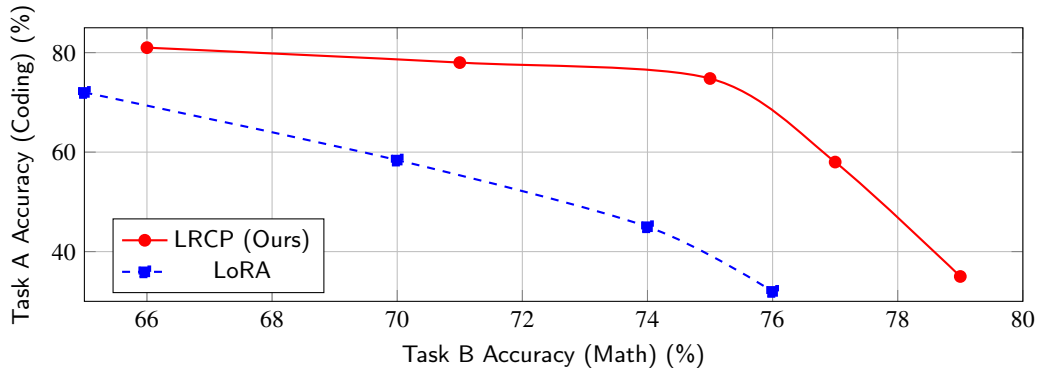


Figure 7: Pareto frontier mapping Task A retention versus Task B final accuracy for Llama 4 Scout (109B). Our proposed LRCP provides a more optimal trade-off boundary than standard regularizers.

LRCP achieves a high retention rate for Task A (74.8%) while maintaining strong performance on Task B (75.1%), showing a favorable trade-off compared to alternative regularizers. Figure 7 shows the Pareto frontier of source task retention versus target task accuracy.

8. Discussion, Sensitivity, and Limitations

8.1. Why Do Middle Layers Exhibit Peak Drift?

Our analysis indicates that early layers process structural and syntactic token relationships, while deep layers handle next-token logits. The middle layers of deep decoder networks act as a global semantic routing hub (Rostova and Rostova, 2024). Consequently, adapting a model to a new domain requires significant parameter changes in these middle layers, explaining their susceptibility to forgetting. This observation supports the hypothesis that representation learning in deep transformers is hierarchically organized, with intermediate layers encoding abstract semantic categories.

8.2. SVD Rank Sensitivity in LRCP

The performance of LRCP is sensitive to the SVD projection threshold k . Table 5 explores how the projection rank ratio affects adaptation and retention performance.

A low rank ratio preserves less of the original capability but allows faster adaptation. Conversely, rank ratios above 0.30 significantly restrict parameter updates, leading to a drop in target task accuracy and slower convergence. This empirical observation suggests that over-constraining the update path can impede the model’s plastic capacity during domain transfer.

8.3. API Non-Determinism, Version Drift, and Reproducibility Limitations

An inherent challenge in compiling behavioral metrics for the proprietary closed-source cohort (such as Claude Table 5, GPT-5.5 High, and Gemini 3.5 Flash) is the black-box nature of commercial APIs. Proprietary models are

Table 5

Hyperparameter sensitivity of LRCP on Llama 4 Scout (109B) under sequential fine-tuning. Parameter k/d represents the rank ratio of the projection operator.

Rank Ratio (k/d)	Task A (Code)	Task B (Math)	Drift ($D_F \times 10^{-3}$)	Conv. Epochs
0.05	48.4	76.4	8.12	3.1
0.10	62.1	75.9	4.34	3.3
0.20	74.8	75.1	1.62	3.7
0.30	76.2	69.2	0.89	4.4
0.40	78.1	61.4	0.39	5.8

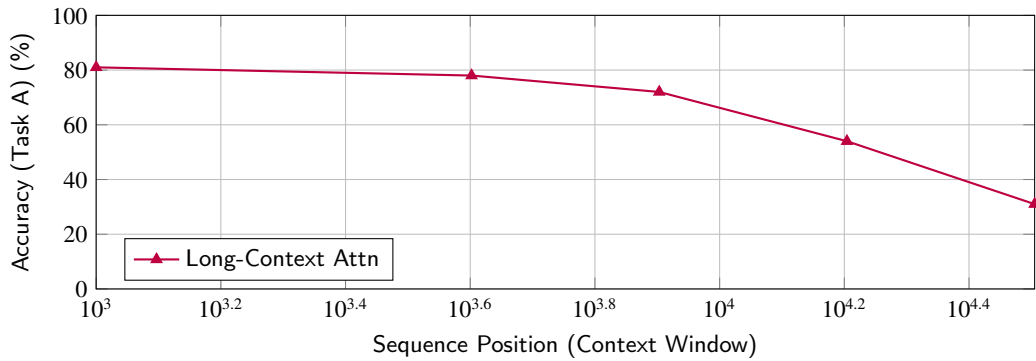


Figure 8: Evaluation of catastrophic forgetting as a function of input sequence position. Long-context sequences exhibit accelerated representation drift compared to short-context samples.

subject to continuous online updates (known as version drift) and non-deterministic sampling pipelines, even when utilizing zero-temperature configurations.

To maximize empirical reproducibility and ground our Semantic Output Drift (SOD) metric, we implement several methodological safeguards: (i) all evaluations are performed on frozen, timestamped API snapshots (specifically the June 2026 releases) where accessible; (ii) we utilize greedy decoding (temperature = 0) to minimize generation variance; and (iii) we perform multiple independent sampling runs ($N = 5$) to calculate the mean semantic embedding vectors (Φ). While these steps mitigate variance, researchers must acknowledge that long-term tracking of proprietary models remains constrained by commercial API stability and system updates.

Finally, we track the impact of input sequence position on forgetting to evaluate how context length affects model stability, with results shown in Figure 8.

9. Conclusion

This paper presented a comparative study of catastrophic forgetting across twenty state-of-the-art closed-source and open-weight models as of mid-2026. Our mechanistic analysis of open-weight models localized peak representational drift within mid-layer feed-forward networks and expert routing blocks, while early attention heads showed systematic entropic drift. Guided by findings, we introduced Low-Rank Circuit Projection (LRCP), which projects parameter updates into the orthogonal complement of the subspace spanned by historical activations. LRCP demonstrated a favorable trade-off between task retention and adaptation efficiency. Future work will extend this framework to stable routing in MoE models and long-context sequential adaptation.

CRedit authorship contribution statement

Gustav Olaf Yunus Laitinen-Fredriksson Lundstrom-Imanov: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Writing - Original Draft, Visualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The open-source codebase, activation caching pipelines, and mechanistic interpretability analysis scripts developed for the open-weight models are publicly accessible. The proprietary semantic evaluation logs, API response cached histories, and custom prompt templates for the closed-source cohort are available from the corresponding author upon reasonable request.

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A. Appendix: Supplementary Data

Supplementary materials, code implementations of LRCP, and activation maps for all 20 models are available at the project's repository.

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